Bipul Biswas

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.90
Kendriya Vidyalaya Bilaspur	Class XII (Central Board of Secondary Education)	2018	90.00%
Kendriya Vidyalaya Bilaspur	Class X (Central Board of Secondary Education)	2016	10.00%

MTECH THESIS

Silicon-on-Insulator Microring Resonators: Integrated Cavity Design, Fabrication Optimization, and Active Thermal Drift Compensation via Dithering

Guide: Dr. Naresh Kumar Emani

- **Design:** Modeled mode profiles, coupling regimes, and Q-factors using ANSYS Lumerical FDE and 3D FDTD.
- **Fabrication:** Optimized cleanroom lithography, RIE etch recipes, and thin-film metallization to minimize optical loss.
- **Control:** Designed an active thermal dithering feedback loop to automatically lock resonance and eliminate thermal drift.

SCHOLASTIC ACHIEVEMENTS

Indo-Taiwan Collaborative Workforce Development Program, Hsinchu, Taiwan

Selected as a participant in a semiconductor short-term training program organized by the Tze-Chiang Foundation of Science & Technology (TCFST) in collaboration with IIT Hyderabad (Mar 2025) to gain exposure to semiconductor industry practices in the Taiwanese core fab ecosystem.

SKILLS

Languages: System Verilog, Verilog, C, Python.

Skills: ASIC Design, Physical Design, FPGA Prototyping, Hardware Software Interface

Protocols: AXI Stream, AXI Lite

Tools: Xilinx Vivado, Vitis, Cadence Virtuoso, Cadence Genus, Innovus & Xcelium, LtSpice.

PROJECTS

32-Bit RISC-V Single-Cycle Core Design*(Tools used : SystemVerilog, Xilinx Vivado)*

- The design of a single-cycle 32-bit RISC-V (RV32I) processor realized in the form of SystemVerilog is presented. This core supports R, I, S, B, U, and J-type instructions.
- Verilog testbenches were developed to verify the core functionality with assembly benchmark programs such as array max-search and Fibonacci number generation.
- The design correctness was established, and the memory write-backs were verified to be cycle-accurate using the Xilinx Vivado behavioral simulations and waveform analysis.

AXI4-Lite Memory Slave IP Verification*(Tools used : SystemVerilog, Xilinx Vivado)*

- Validated independent 32-bit read/write channel logic for an AXI4-Lite memory slave IP against ARM protocol compliance, by building a mailbox-based SystemVerilog testbench with generator, driver, monitor, and scoreboard

components and FSM-based interface design.

1D Semiconductor Heterojunction Device Simulator*(Tool used: Python / NumPy / Matplotlib)*

- Developed a 1D simulator using Newton-Raphson to solve non-linear Poisson equations in semiconductor heterojunctions.
- Modelled parameter discontinuities and boundary conditions across multi-material interfaces.
- Simulated spatial profiles for potential, electric field, charge density, and carrier concentrations under bias.

Synchronous FIFO Design & Verification *(Tools used: SystemVerilog, Xilinx Vivado)*

- This project realizes a 16-deep 8-bit synchronous FIFO in Verilog using pointer based circular buffer logic and combinatorial full/empty flags.
- A class-based SystemVerilog testbench (generator, driver, monitor, scoreboard) is developed using mailboxes and events to randomly drive the read/write transactions.
- The FIFO is verified using a queue-based scoreboard reference model resulting in 90+ randomized transactions without any data mismatches and waveform analysis.

RELEVANT COURSES

Digital Domain: Digital IC design, Introduction to VLSI design, Computer architecture, Neuromorphic devices for AI/ML, Digital IC design Lab

Analog domain: Analog IC design, Analog IC design Lab.

Technology & devices: VLSI Technology, Semiconductor device modeling, High speed semiconductor optoelectronics, Modeling of nanoscale MOSFETs

POSITIONS OF RESPONSIBILITY

Teaching Assistant, High-Speed Semiconductor Optoelectronics (IIT Hyderabad):

Led problem-solving sessions for 15 students, boosting assignment completion rates.

Teaching Assistant, NPTEL -- Introduction to Semiconductor Devices:

Resolved device physics queries for 100+ national learners, reinforcing core semiconductor fundamentals.

EXTRACURRICULAR

Student Chapter Leadership: Spearheaded design initiatives, member engagement drives, and technical facility coordination for the institute's SPIE Student Chapter.